

Tuning in and tuning out: media multitasking among young consumers

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- *While commercial media are increasingly attended to in multitasking contexts, advertising research is dominated by sequential marketing communication models where the consumer is a passive receiver of one type of media at a time. This study examines media multitasking behaviors and experiences among young consumers. Media multitasking is the practice of participating in multiple exposures to two or more commercial media forms at a single point in time, including traditional, online, social, and entertainment media. Prior research argues that multitasking results in diminished comprehension and performance. Through a qualitative approach with a sample of 64 young consumers involving interviews and collages, this study highlights the paradoxical experiences of media multitasking. The authors develop a conceptual model that explains the personal benefits and costs associated with media multitasking, and proposes its impact on consumers' motivation, ability and opportunity (the MAO model) to process commercial media content. Further, the study suggests an inverted-U relationship between media multitasking and the MAO model and proposes four behavioral coping moderators to this relationship.*

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"Multitasking is a stressful situation because you are doing so many things, but at the same time, you are kind of efficient. You know what is going on, but there's so much going on at once that you are not taking it all in. It's pretty

chaotic. . . people are picking up the phone, chatting on the Internet, watching TV, reading something. You become bombarded with information and ads. But you have to be efficient in order to manage it all." (Bill, Interview, 22 years old)

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This quote from our research highlights a common media phenomenon we refer to as media multitasking—the simultaneous consumption of two or more commercial media vehicles or content. Media consumption in such contexts is often cognitively challenging,

producing a complex and often stressful media experience that can influence individuals' motivation, ability, and opportunity (MAO) to attend to and process commercial messages, including advertising. The quote also hints at the coping strategies consumers have developed to become more effective in multitasking contexts. The media multitasking context raises questions regarding our understanding of consumer engagement and comprehension with commercial media and content. The purpose of this study is to examine media multitasking and its impact on MAO to process advertisements in a multitasking media environment. More specifically, we (1) determine the costs and benefits of multitasking; (2) identify multitasking coping strategies; and (3) propose the impact of media multitasking on the MAO model.

Industry research demonstrates that commercial content is increasingly attended to in multitasking situations (25–30% of total media time), and the more media an individual consumes, the more likely she will consume several media types at once (Pilotta and Schultz, 2005; Foehr, 2006). Nielsen Media estimates that almost one-third of all household Internet activity takes place while users are watching television. At the same time, media consumption is becoming increasingly social, as individuals instant message, email, text or talk on the phone while at the same time tuning into commercial media such as television.

Media multitasking is a phenomenon born from the myriad interactive media platforms easily accessible to consumers. Contemporary media include multiple forms of information presentation, such as television newscasts displaying multiple messages on one screen that enable viewers to access several different news items simultaneously (e.g. Chowdhury *et al.*, 2007). Consumers are no longer passive media spectators, but interact with media in co-production settings. Moreover, time compression is a commonplace occurrence in media consumption; research has shown that U.S. consumers on average undertake 43-h of activities through multitasking in a

24-h day. Sixteen of these multitasking hours consist of interaction with media and technology (Research and Markets, 2006).

While a predominance of media consumption has shifted to multitasking situations, advertising research has continued to examine media consumption within a sequential marketing communication model where the consumer is a passive receiver of one type of media at a time. Commenting on the state of research on this issue, Schultz (2002, p.6) notes that "What we really need today is a new approach to media planning, one that recognizes consumers' increasing ability to multitask." Prior research has investigated the influence of media interactions and synergy across multiple forms of media (Edell and Keller, 1989; Naik and Raman, 2003). However, these studies have not involved simultaneous access to and consumption of media. One exception is the study of the simultaneous presentations of advertising and television programming (Chowdhury *et al.*, 2007) that suggests multitasking may have a diminishing impact on strong messages, but not for relatively weak messages. Our study differs from the existing work in media multitasking (Pilotta and Schultz, 2005; Chowdhury *et al.*, 2007) in two ways. First, we take a more in-depth understanding of the phenomenon through qualitative research. Second, while prior work looks at multitasking as a negative and destructive activity, we identify not only its costs, but also its benefits as well as the coping strategies that make consumers more effective at media multitasking.

We aim to advance our understanding of media consumption through the study of this important phenomenon of media multitasking. We provide a conceptualization of media multitasking and examine it through a qualitative study with a sample of Gen Y consumers, as this phenomenon is commonplace within the Gen Y culture. We develop an integrated model of media multitasking that identifies multitasking motivations and coping strategies, and then propose its impact on the MAO to process advertisements. Prior research argues that multitasking results in diminished

comprehension and performance. However, our findings argue for a more complex discussion of the impact of media multitasking on consumer comprehension of commercial messages. Instead of a negative linear relationship between media multitasking and consumers' MAO to process commercial content that underline prior research, we propose an inverted-U relationship. Additionally, we identify four behavioral coping strategies and propose their moderating effect on the relationship between media multitasking and the MAO model. Overall, this research contributes to marketing communication theory and practice by studying the ways consumers cope with the abundance of media and remain able and motivated to attend to, process, and comprehend commercial content from multiple sources.

Background literature

Multitasking behavior

Multitasking has been described as the ability to accomplish "multiple task goals in the same general time period by engaging in frequent switches between individual tasks" (Konig *et al.*, 2005, p. 243). Consumer research has associated multitasking with polychronic time use, which has found that consumers intentionally multitask to address multiple role demands (Kaufman *et al.*, 1991).

The dominant perspective in multitasking comes from cognitive psychology and argues that because of limited cognitive resources, people do not have the ability to effectively multitask. Thus, multitasking behaviors may inhibit or diminish task performance. As suggested by resource allocation theory (Kanfer and Ackerman, 1989), the human brain has finite cognitive abilities to attend to multiple tasks simultaneously. Coordination among tasks is diminished so as to delay action and create gaps in task completion (Rubenstein *et al.*, 2001). Multitasking can be especially inefficient during complex or non-familiar tasks, where individuals are

required to expand significant attentional resources (DeShon *et al.*, 1996). Overall, cognitive psychology argues that consumers are less effective during multitasking. We examine this view in the context of media consumption.

Media multitasking and its impact on MAO model

We introduce media multitasking as a specific case of multitasking behavior. Media multitasking has been described as the simultaneous consumption of various forms of media (Pilotta *et al.*, 2004). Drawing upon this, we conceptualize media multitasking as the practice of participating in multiple exposures to two or more commercial media forms at a single point in time, including traditional, online, and social media. Media multitasking can occur during interactions with multiple communication media, such as watching a program on TV, listening to one's iPod while chatting with friends on Facebook, and shopping online at Amazon. It can also take place around one medium, such as a personal computer, where one is shopping online while reading the online news and instant messaging with friends.

Media multitasking has attracted the attention of advertisers because of the increased challenges of drawing attention and engaging young consumers (Schultz, 2002; Wallis, 2006). Multitasking represents the context in which advertising is processed. Thus, we consider its impact on the MAO model with respect to processing media content. The MAO model represents the antecedent conditions of the elaboration likelihood model (ELM) where variations in MAO influence levels of consumers' information processing (Petty and Cacioppo, 1986; MacInnis and Jaworski, 1989; MacInnis *et al.*, 1991). The ELM suggests that elaboration of advertising messages is more likely when consumers possess the motivation, ability, and opportunity to process the exposed information. *Motivation* refers to consumers' willingness to invest cognitive

effort in processing commercial content during multitasking and relates to consumer involvement with the subject matter (MacInnis *et al.*, 1991). Consumers with high motivation would be willing to allocate processing resources to commercial content during media multitasking. *Ability* refers to the consumer's accessibility and skills needed to decode information (MacInnis *et al.*, 1991). High ability refers to the presence and accessibility of prior knowledge. *Opportunity* is to the extent to which distractions or limited exposure time affect a consumer's attention to commercial content (MacInnis *et al.*, 1991)

Cognitive psychology research would suggest that media multitasking can diminish consumers' MAO to process commercial content because the consumer is exposed to more stimuli and greater demands on attentional and cognitive resources compared to a single exposure context. Thus, media multitasking can diminish the opportunity to process information since consumers face several distractions that draw their attention from one stimulus to another (MacInnis and Jaworski, 1989). Further, greater effort is needed to process information during multitasking since it may lead to interfering learning effects and reduced accessibility of knowledge structures (Edell and Keller, 1989). Multitasking may diminish consumer motivation since it provides several options that compete for attention, reducing the motivation to process content (Kanfer and Ackerman, 1989). Overall, the cognitive psychology perspective argues for a negative, linear relationship between media multitasking and the MAO model.

However, evidence shows that consumers may develop skills to cope with multitasking. We examine media multitasking from the perspective of the consumer and draw upon paradoxes of technology theory (Mick and Fournier, 1998; Johnson *et al.*, 2008) to guide our understanding of consumer costs and benefits from media multitasking. This theory would suggest that while consumer interaction with media can be challenging, these costs co-

exist with benefits that consumers may derive from multitasking. Consumers learn to cope with this paradoxical experience. Consumer socialization with media and technology may make media multitasking activities automated and ritualistic. Consumers may become more effective processors while multitasking when they use technology to complement or supplement other media (Sinan *et al.*, 2007). Further, consumers have developed new skills for attending to and decoding media content and are able to compartmentalize this content (Dresner and Barak, 2006; Jenkins, 2006). Moreover, media multitasking allows processing opportunities to be exposed within the same commercial message repeatedly, across several screens, and research on interactive advertising finds that communications with interactive elements result in increased encoding and comprehension (Marcias, 2003). Therefore, an alternative perspective emerges arguing that some consumers may be more adept at multitasking because their socio-cultural milieu may have enhanced their motivation and ability to process media content in multitasking contexts. We build on this perspective and take a more balanced look at media multitasking, arguing that its impact on consumer MAO can also be positive and that consumers develop behavioral strategies to deal with the costs of multitasking.

Method

This is an exploratory study aimed at understanding the concept of media multitasking. We employ a qualitative approach, which is deemed appropriate for exploratory studies (Peters *et al.*, 2007). Our approach also enables us to bring the consumer experience to the study of multitasking (Belk *et al.*, 1988), which has predominantly been cognitive and has largely ignored consumer experiences and practices.

While media multitasking is common across consumer groups in the U.S., it is a particularly entrenched practice among the youth or

Generation Y (Gen Y) consumer (Foehr, 2006).¹ This is why we selected Gen Y individuals to study this phenomenon. In a nation-wide survey study reported by *Time* magazine, Americans ages 8 to 18 are spending 8.5 h a day using electronic media (Wallis, 2006). It is common these days for young consumers to be sitting in the living room doing their homework with the TV on while talking or text messaging on their cell phone, listening to their iPod, and checking out their Facebook or MySpace pages, all at the same time. Gen Y lives in a world surrounded by, even defined by, communication technologies in which work, leisure, socialization, and personal self-development are mediated by exposure to various media technology (Tapscott, 1998). We deemed a convenience sample of undergraduate students as appropriate for examining media multitasking among this age cohort. We sampled 64 American undergraduate students at an urban, northeastern university in the U.S. The sample has an equal gender distribution (46% female and 54% males) that falls into the 20–23 year old age category. Prior to the study, informants reported their media ownership and usage in a brief survey. All informants reported owning a cell-phone; 97% of informants owned and used an iPod daily; and 94% of informants were active on social networking sites (e.g., Facebook). Informants reported that they typically multitasked online and while watching television.

Data for the study were collected through semi-structured interviews and collages. Researchers conducted semi-structured interviews with 30 of the 64 students ranging in length from 45 to 90 min (McCracken, 1988). The interviews followed a pre-established interview guide that consisted of questions regarding students' typical daily media consumption, the motivations behind their multitasking, and their overall experiences with multitasking. In addition to the interviews, we asked the other 34 students to express their

media multitasking experiences through the development of pictorial collages accompanied by one-page written essays describing the visuals represented (Belk *et al.*, 2003). Students were instructed to construct their own personal collages using a variety of popular magazines. They were encouraged to freely express their feelings and associations regarding multitasking with media. Following the completion of their collage, they wrote an essay of their interpretations of the collage. The purpose of the collage generation exercise was to supplement the interview findings with an approach that allowed for pictorial interpretation in order to elicit the media multitasking experiences, behaviors, and meanings that might not arise during self-reporting interviews (Coulter, 2007).

Qualitative analysis followed separately for each type of data collection, and the findings were integrated at the end. The interview transcripts were analyzed and coded by the researchers following proper procedures of qualitative data analysis (Strauss and Corbin, 1990). At first, the authors carefully read the data to familiarize themselves with the informants' experiences related to media consumption. Within-case analysis was conducted to identify for each informant: (a) media multitasking behaviors and experience, (b) personal costs and benefits from media multitasking, and (c) the impact of multitasking on MAO related to processing commercial content. Cross-case analysis of the data aimed at identifying common themes across informants. Transcript coding was guided by the study's research questions as well as the theoretical background of the study, which included media paradoxes of technology theory, multitasking behavior theory, and resource allocation theory. Data analysis and interpretation were both deductive and inductive: the expectations from the literature review and prior multitasking work served as the preliminary themes that guided our initial data analysis. More specifically, the eight paradoxes of technology (Mick and Fournier, 1998) guided the analysis of multitasking costs and benefits; four of the paradoxes were the same

¹Gen Y (also known as "the Millennials") refers to a specific cohort of individuals born from 1983 to 1997 (Solomon 2008).

in the multitasking context as in the technology contexts they study. The coping strategies were the new themes that emerged from the data. Next, we analyzed the collages based on the accompanying essays in order to gain a deeper understanding of the media multitasking experience. The essays not only explained the collages, but also generated useful metaphors for further data interpretation. The findings from the collages were consistent with the interview findings and provided richer details of media multitasking behaviors and experiences. The findings from the analysis of the interviews and collages are discussed next.

Findings

Based on the data, we develop an integrated conceptual model of media multitasking (see **Figure 1**) that examines the costs and benefits associated with media multitasking and suggests five propositions regarding the impact of multitasking in consumers' motivation, ability, and opportunity to process commercial content. The model emerged from the data and is intended to provide an interpretive framework to organize and enhance the discussion that follows. This section introduces the media multitasking costs and benefits, as well as behavioral coping strategies that moderate the relationship between media multitasking and the MAO model.

Media multitasking costs and benefits

One goal of this study is to identify consumer costs and benefits from media multitasking. Consistent with the cognitive psychology perspective on multitasking, our data suggest that media multitasking is challenging for informants' MAO of commercial content. However, we also find that informants derive benefits from media multitasking that may at the same time enhance their MAO. Thus, the media multitasking experience can be characterized as paradoxical and parallels a subset of consumer paradoxes of technology developed by Mick and Fournier (1998): control/chaos, efficient/inefficient, engagement/disengagement, assimilation/isolation, and freedom/enslavement. Regarding the last two paradoxes, isolation and freedom did not emerge clearly in the data. We explain each of the costs and benefits below and provide illustrative quotes for each in **Table 1**.

Costs of media multitasking

Our data suggest that media multitasking has four negative consequences for consumers: inefficiency, chaos, disengagement, and enslavement. Consistent with prior research on multitasking behaviors, our data indicate that media multitasking has a negative impact on MAO. More specifically, media multitasking creates a chaotic and inefficient media environment, in effect reducing consumers' opportunity and ability to process advertising content.

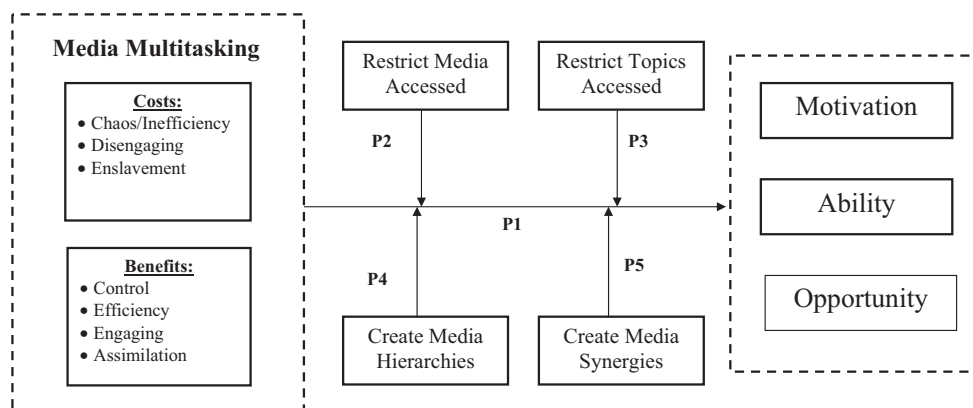


Figure 1. Model of Media Multitasking Impact on MAO Model.

Table 1. Consumer costs and benefits from media multitasking

<i>Motivations</i>	<i>Conceptualization</i>	<i>Illustrative quotes²</i>
Inefficiency	Media multitasking can lead to distractions, disorientation, and procrastination of tasks.	"When I'm watching TV and talking on the phone, I don't pay attention to the phone call, hardly at all... But at the same time after I hang up I'm still behind in the show or game or whatever. Although I'm doing two things, afterwards it's almost like neither of them happened." (David, Interview, 21 years old)
Chaos	Media multitasking can lead to an experience of disorder and upheaval.	"The image of the woman's face is covered in different colors; each layer represents how my attention is divided between tasks and the fact that my attention span and dedication is small and divided. The fact that the woman is looking down expresses the fact that I often feel guilty about multitasking because I am not being honest toward the task at hand" (Laura, Collage, 23 years old)
Disengagement	Media multitasking can reduce consumers' involvement with processing a particular message.	"Dazed; confused; not paying attention to anything really. Blank mindedness or numb. When I'm watching TV, I'm always doing something else. So I don't really pay attention to the TV. But at the same time I'm also not really paying attention to the other thing I'm doing." (Rich, Interview, 21 years old)
Enslavement	Media multitasking can be an addictive experience.	"I think I am probably A.D.D. undiagnosed. You know, the cell phone buzzes and you get that text message and you just want to jump into it. And then you get an IM and you jump away from whatever you are doing and you got to focus there, losing track of the TV program that might be on." (Brandon, Interview, 20 years old)
Control	Media multitasking can enable consumers to effectively filter and process marketing communication content that they are interested in.	"When I am multitasking I feel like I am the operator of a mission impossible, in control. I have all the information at my reach and disposal... I feel like I have a handle on all of the different media types and am using them to supplement each other. I am interpreting all of the media at once and am obtaining some solid information" (Alex, Collage 3, 20 years old)
Efficiency	Media multitasking can enable consumers to carry out tasks and process content faster.	"When I am multitasking it is exciting and fast paced; you have the pulse of the world. It's a lot of stimuli at the same time. For instance, I got the TV on mute with the closed captioning on, and I'm listening to my favorite band, and I happen to turn around and a flash from the TV catches my eye and I see my favorite band is coming to the area and then I go online to check for tickets. There is no time lost, you have the technology at your hands to make all your dreams come true. Now you're satisfied. Instant gratification!" (Amanda, Interview, 23 years old)
Engagement	Consumption of media through multitasking constitutes a hedonic experience.	"All pictures convey emotions of happiness or up-beat feelings. NOISE and eyes illustrate the method of consumption between mediums. I usually use computers, television and music to relax and entertain myself. This might seem counterintuitive (with the words NOISE), but my eyes and ears almost need the different mediums to stay intrigued." (Steve, Collage 1, 22 years old)
Assimilation	Media multitasking can enable individuals to connect with friends and family as well as with cultural influences.	"One major advantage of media multitasking for me is the feeling of being 'connected'. If I am watching the news on TV while talking on my cell phone and checking out Myspace.com, I feel in synch with what is happening in today's world, in addition to the person with whom I am having a text message conversation and the numerous people with whom I have relations on Myspace." (Tom, Interview, 22 years old)

²Note: We found quotes from many informants that support each of these themes; however, we present here only one illustrative quote for each theme.

Informants reported taking longer times to process content during multitasking. Finally, disengagement with particular media content may reduce consumers' motivations to process advertising content, and the enslavement experience may further challenge the opportunity to process content.

Inefficiency

The data suggest that media multitasking was an inefficient media experience. Informants described media multitasking as distracting, leading to procrastination of tasks and diminished learning. It was referred to as a distraction of the informant's attention from one stimulus to another. Thus, media multitasking may reduce the opportunity to elaborate on commercial content. Informants reported that they paid less attention to specific tasks and needed much more time to decode media content during multitasking since they were continuously dividing their attention among various media (as illustrated by David's quote, Table 1) (Edell and Keller, 1989; MacInnis and Jaworski, 1989).

Chaos

Multitasking with media can also lead to an experience of disorder and upheaval. In Laura's essay (see illustrative quote, Table 1 and collage in Exhibit 2), she characterizes her media multitasking experience as accompanied by divided attention and lessened opportunity to process compared with single-message processing. Informants were not only aware of their inability to process content effectively during multitasking, but they also expressed negative emotions, such as stress and guilt associated with the chaotic experience of media multitasking.

Disengagement

Mick and Fournier (1998) found that technology can lead to disconnection, disruption, and passivity because it automates action. In our

context, we found that media multitasking led some informants to feel less engaged with processing content from their primary media because multitasking diminished their opportunity and ability to process message content. In general, we found media multitasking may reduce consumers' involvement with a particular message, and as a result, reduce their willingness to invest cognitive resources to attend to the commercial message. Informants indicated that media multitasking fulfilled their need for constant stimulation; however, they often became disengaged and disconnected from attending to any particular media task, as illustrated by a quote from Rich in Table 1.

Enslavement

Another negative consequence of media multitasking behavior related to the addictive nature of media consumption was expressed in collages through images of police handcuffs, claims such as "*can't be without you*," and images of pills and chocolates (see Exhibit 2). Informants talked of their dependency on media and technology as well as their need to multitask in order to manage the vast array of media available. Many informants expressed a general experience of "attention deficit disorder" or A.D.D in their lives, characterized by low attention and a need for media stimulation attributed to their media multitasking behaviors (as illustrated by Brandon's quote, Table 1). Thus, the experience of enslavement associated with media multitasking may challenge consumers' opportunity to process commercial content (MacInnis and Jaworski, 1989).

Benefits of media multitasking

However, we also find that the costs associated with media multitasking co-exist with four benefits of multitasking with media. Our informants multitasked with media because (1) multitasking provided them a heightened sense of control over their media consumption, and (2) multitasking helped them

perform more efficiently. Thus, media multitasking may also enhance consumers' ability and opportunity to process commercial messages. Additionally, multitasking made them feel more (3) engaged with the media consumption process and (4) also helped them experience a greater level of assimilation within their social network. In this way, media multitasking may increase consumers' motivation to process commercial content. Our data show that informants often engaged in some level of media multitasking to enhance their involvement with media, to increase their opportunity to decode media content, and to enhance their ability to process media content. These results argue that some level of media multitasking may indeed have a positive impact on MAO.

Control

Our informants expressed that they multitasked because it enabled them to remain in control of their media consumption. Control was manifested in two ways. First, media multitasking enabled them to more effectively filter and process marketing communications content that they were interested in. One informant expressed the use of media under media multitasking as a "*One-Man Band*" to express the active role he plays in the communication process during multitasking. Informants felt as if they were in control of their media consumption as they determined what media content they were going to attend to—in what way, when, and how. Second, informants sought to gain necessary cultural and informational capital through media multitasking, and as such they sought to exert control over this process. Exhibit 3 illustrates these two aspects of control. Through media multitasking, Alex remains in control of his media consumption, which in turn enables him to become a competent media multitasker (see illustrative quote by Alex, Table 1). Consumers also tend to be more involved and interested in the content because they have sought it out (Macias, 2003). In other

words, consumers tend to engage in greater elaboration of communication processes over which they have control.

Efficiency

Similar to general technology use, we find that media multitasking enabled our informants to be more efficient because it involved less effort or time (Mick and Fournier, 1998; Johnson *et al.*, 2008). Prior research on multitasking in the workplace has suggested that people engage in multitasking because they become more efficient in carrying out job-related tasks (Reinsch *et al.*, 2008). We found that media multitasking results in "greater convenience," "saving time," "constant connectivity," and "greater access to information". During media multitasking, an individual may be exposed to similar advertisements across several screens, and this may enhance their ability and opportunity to process the advertisement. Multitasking enabled them to participate simultaneously in multiple worlds as illustrated by Amanda in Table 1.

Engagement

Media multitasking was also viewed as a hedonic experience—fun, enjoyable, and an escape from reality. Similar to entertainment gratifications derived from the use of a single medium (Peters *et al.*, 2007), our informants multitasked with media to "kill time," to be entertained, or to relax. However, compared to engagement with a single communications medium, media multitasking may be experienced as relatively more engaging through exposure to multiple sensory stimuli (see illustrative quote from Steve, Table 1). Multitasking enabled informants to become highly engaged in the process of media consumption, thus increasing their motivation to process commercial content. Because it was engaging and interactive, multitasking activity often produced an experience of being productive. One consequence of the digital generation may be an experience of boredom or

disengagement while not “in action” with respect to media access. Consumption of various media simultaneously was not only viewed as fun but also as productive. One informant commented that “*multitasking doesn’t feel like work, which is probably the reason I do it so much!*” Research on Gen Y consumers suggests that they need to enjoy what they are doing (NAS Insights, 2006), and multitasking enables them to make accessing and processing media content fun. Therefore, we can infer that some level of multitasking stimulation with respect to media is needed to motivate these young consumers to process commercial messages.

Assimilation

Mick and Fournier (1998) find that technology can be used as a social mediating mechanism helping people to assimilate within social contexts. Similar to the use of technology, media multitasking enables individuals to connect with friends and family as well as with surrounding cultural influences. The collages indicated that a benefit of media multitasking was the connectedness it enables. In the illustrative quote, Tom (see Table 1) feels a sense of connectedness through media multitasking, which in turn may enhance his media engagement and motivation to process commercial content. The theme of consumer assimilation through media multitasking was seen throughout our data, and we found no evidence of consumers feeling isolated through media multitasking. Rather, informants were motivated socially to use various forms of online and wireless media to strengthen social networks.

In summary, our data suggest that informants attend to commercial content in multitasking contexts because multitasking enables consumers to be in control, to be efficient, and engaged, and results in assimilation. Since we find that the benefits co-exist next to the costs of media multitasking, instead of a negative linear relationship between media multitasking and MAO, we propose an inverted-U

relationship suggesting that some media multitasking activity is often necessary and may have a positive impact on MOA model.

P1: *Media multitasking has an inverted U-relationship with consumers’ motivation, ability and opportunity to process commercial media content.*

Coping with media multitasking

Another goal of this study is to examine the ways consumers cope with media multitasking and how these coping mechanisms relate to the MAO model. Informants often consumed media strategically in order to cope with the negative impact of media multitasking. Four coping strategies emerged from the data: (1) restricting the number of media platforms accessed; (2) restricting the number of topics accessed; (3) creating media hierarchies; and (4) creating media synergies. The first two avoidance strategies seek to restrict the amount of multitasking during media consumption, thus diminishing the negative impact of multitasking on processing of media content. The last two coping strategies have developed over time as informants become more adept at media multitasking through skill and competency development. Based on these coping strategies, we develop four propositions regarding the way these coping strategies moderate the relationship between media multitasking activity and the MAO model. We explain each of the coping strategy below and provide illustrative quotes for each in Table 1.

Restrict media accessed

The first coping strategy aims to restrict the number of media used at one single point in time. Media devices differ in terms of design and each places unique demands on consumers’ opportunity and ability to process content. For example, a media multitasking situation where one is watching TV and shopping online while listening to the radio

involves three different media, each engaging different sensors at the same time. Switching among these three media requires greater cognitive resources than does switching within one device (Rubenstein *et al.*, 2001). Therefore, consumers' ability and opportunity to process advertising content may be diminished. Some informants attempted to control media access by setting spatial and physical boundaries, or by simply switching off interactive media that were more detrimental to attention, as illustrated by Amy in **Table 2**. Overall, we suggest that multitasking around one media device, such as a PC, may challenge consumers' ability and opportunity to process commercial content less than it would multitasking around two or more media devices. As such, we propose a positive moderation effect:

P2: *Restricting the total number of media involved in multitasking will positively moderate the relationships between media multitasking and the consumer's ability and opportunity to process commercial media content.*

Restrict topics accessed

The second coping strategy involved restricting the number of topics being processed at one time during media consumption. Our informants often sought to complement and add depth to their media experience by focusing on one topical area. For instance, some informants spoke of watching a sporting event on television and checking concurrent game statistics online while at the same time participating in a related online fantasy sports league. When informants switched between various media or vehicles that were centered on one particular topic or task, they experienced a greater degree of control, were more engaged with the topic of interest, and were motivated to pay attention to commercial content related to the central topic. Additionally, informants had more opportunities across multiple screens to be exposed to and process the commercial content related to the topic of interest. Informants reported that they learned

to compartmentalize their attention to, and processing of, multiple media around a singular topic. As Matt states in **Table 2**, multitasking around a topic enables him to avoid distractions from other content, and this may enhance his motivation, ability and opportunity to process commercial content. Overall, our data suggest that the restriction of topics during multitasking may reduce the intensity of media multitasking that the consumer deals with and as a result, may also enhance their ability to process commercial content. Our data thus suggest another moderator to the relationship between media multitasking and MAO model:

P3: *Restricting the number of topics during multitasking will positively moderate the relationship between media multitasking and the consumer's motivation, ability, and opportunity to process commercial media content.*

Create media hierarchies

The third coping strategy involves the creation of media hierarchies during media multitasking: traditional media were often consumed as background media, whereas new media were mostly consumed as foreground media. Digital, interactive media were portrayed as more intrusive yet engaging, demanding greater consumer action, and as such took on a foreground role. "*Traditional media do not require my full attention*" and are often consumed as background media. Informants watched television with the sound off, yet, television continued to be a central part of their media hierarchies. A medium was considered background when its use was described as a secondary task. Informants expressed that they paid less attention to background media and used it to fill the voids in their social environment. Yet, both media platforms (television and the Internet) were attended to depending on the informant's setting and goal(s). Thus, media hierarchies are seen to be dynamic (cf. O'Guinn and Shrum, 1997). Such media hierarchies operated as a

Table 2. Consumer coping strategies with media multitasking

Coping strategies	Conceptualization	Illustrative quotes ³
Restrict media accessed	Restrict the number of media used at one point in time during media consumption.	"Well, I'd have to set boundaries so I wouldn't multitask when I had to do homework. Just don't have TV on when I have to do homework and if I have to write a paper don't have your AOL buddy list open." (Amy, Interview, 22 years old)
Restrict topics accessed	Restrict number of topics being processed at one point in time during media consumption	"I only pay attention to what I am actively seeking. I have become adept at tuning out things I don't want or need to see and hear. Kind of like a horse with blinders on—I skim and scan material for what I want, and I am effective at that." (Matt, Interview, 21 years-old)
Create media hierarchies	Consume some media as foreground and others as background.	"If I do use two or more media I focus more on one of them. For example, if I am listening to my iPod and reading the <i>Metro</i> , I am usually more focused on the music playing because it is directly in my ears, while I would just be skimming the articles in the paper." (Rachel, 20, Interview, years old)
Create media synergies	Strategically pair media platform to utilize media synergies and reduce the complexity of multitasking behavior.	"I need the background noise; I don't really like straight silence. I am more likely to be distracted and stare off and wonder about things if it is silent. If I am studying and there is no music on then I am more likely to fiddle with my e-mail and AIM...if I have the music on I know I am supposed to be studying. I think I look to have that kind of low-scale distraction and as long as it is there I can focus really well." (Sharon, Interview, 21 years old)

³Note: We found quotes from many informants that support each coping strategy; however, we present here only one illustrative quote for each strategy.

coping strategy in media multitasking contexts because they enabled informants to prioritize and compartmentalize attention and allocation of cognitive resources. Rachel (see quote in Table 2) talked of how her media experience was enhanced by media hierarchies. She attempted to cope with the abundance of available media by prioritizing attention. She paid attention mostly to one medium (her iPod), while she consumed other media (e.g., the newspaper) as background. Our data suggest that certain background media enhanced informants' engagement with media and their motivation to process content in the foreground media.

Overall, consumers strategically create media hierarchies as a way to enable compartmentalization of information and increase the opportunity and ability to process commercial content. Further, consumption of background media is also important to enhance consumer engagement with media activity and, as a result, their motivation to process commercial content. Thus, we propose the following:

P4: *Creation of media hierarchies will positively moderate the relationship between media multitasking and consumers' motivation, ability, and opportunity to process commercial media content from the foreground media.*

Create media synergies

The fourth coping strategy involved the creation of media synergies. We find that our informants engaged in strategic pairing of media platform to utilize media synergies and reduce the complexity of multitasking behavior. Informants demonstrated that they recognized the strengths and limitations of specific media, developing combinations of media that worked together in synergy to create an effective multitasking situation. As shown in Exhibit 1, Steve not only illustrated his knowledge of media synergies, but also that he has developed processing skills that enable effective message processing under multitasking situations: "*There are synergies among all*

media. I feel that during multitasking, many things happen at once, but in my mind they are still easily organized."

We found that informants often would complement resource-dependent media (media that required greater cognitive resources) with resource-independent media (Kanfer and Ackerman, 1996) as a way of being effective multitaskers. Sharon illustrates how she would listen to different types of music depending on which other media or tasks she was involved with (see illustrative quote, Table 2). Depending on the amount of attention required for each respective task, she sought to pair demanding tasks with non-demanding tasks in order to be more efficient in her task accomplishment (e.g., homework). Media multitasking around music was used to enhance her motivation to process content. Our informants' experiences are also supported by the resource allocation theory, which argues that less cognitive and attentional resources are required to carry out tasks that are familiar or automatic for the consumer (Hawkins and Hoch, 1992; Kanfer and Ackerman, 1996). Media multitasking is a daily routine for many. Additionally, informants learned to process resource-independent media in combination with resource-dependent media, increasing their ability and opportunity to process commercial content. Further, this may enhance consumer engagement with more resource-dependent media, leading to greater motivation to process commercial content. Therefore, we propose the following:

P5: *Creation and utilization of media synergies will positively moderate the relationship between media multitasking and consumers' motivation, ability, and opportunity to process commercial media content.*

Discussion

Media multitasking is the way young consumers interact with commercial media. This

presents both challenges to and opportunities for the development of marketing communications and advertising strategy. As consumer segments such as Gen Y become more proficient in decoding media and advertising content in multitasking situations, they also exert more control over what messages they tune into, or tune out of. At the same time, our findings argue that media multitasking may enhance consumers' engagement with and motivation to consume more media over extended periods of time and may provide greater opportunities to target consumers through multiple, different screens.

We develop an integrated model of media multitasking that illustrates multitasking costs and benefits, and suggest its impact on MAO to process commercial content. Our findings highlight a paradoxical experience of media multitasking and identify not only costs, but also benefits from media multitasking behaviors. Rather than a negative linear relationship between media multitasking and the MAO model that underlies prior cognitive psychology research, we propose an inverted-U relationship. We suggest that some level of media multitasking activity heightens a sense of control and efficiency over media consumption and enhances consumers' engagement with and assimilation through media, and as a result, can have a positive impact on the MAO to process commercial media content. We invite future research to empirically investigate the inverted-U relationship proposed here in order to identify the level and types of media multitasking behaviors that allow for a positive impact on the MAO model.

We propose that restricting the total number of media platforms and topics as well as forming strategic media hierarchies will positively moderate the relationship between media multitasking and consumers' MAO to process commercial content. We also propose that the creation of media synergies, such as strategically pairing resource-dependent with resource-independent media tasks, may positively moderate the relationship between media multitasking and consumers' MAO to process commercial media. Our study argues

that marketers should attempt to facilitate consumers' coping behaviors in media multitasking and, in turn, design marketing communication campaigns that consider consumers' strategic multitasking activity.

An example of a campaign that was successful in engaging young consumers was a goal-directed, interactive advertising campaign launched by the sporting goods brand Reebok, called "*Whodunit?*". This campaign attempted to foster viewer control and engagement by providing consumers' goal-directed tasks across multiple media within a common, umbrella narrative. This campaign engaged consumers by having them solve a fictional mystery related to a line of basketball shoes. Mystery cues were embedded in advertisements across multiple media, including TV, print, text-messages, and online. The campaign was successful in terms of driving website traffic, capturing contact information, and promoting outcomes such as consumer engagement, efficiency, control, and assimilation by linking goal-directed tasks across these commonly-accessed multiple media.

Additionally, our findings suggest that advertisers should consider the role that each singular media plays in marketing communication campaigns in multitasking contexts. We found that different media play different roles and elicit different levels of MAO during media multitasking. Thus, companies can target consumers through campaigns (delivered strategically across media platforms) that employ resource-independent media in order to direct consumers to more resource-dependent media to foster consumers' MAO to process the commercial content.

Finally, our findings advance our understanding of multitasking benefits and costs in the context of media consumption by applying the paradoxes perspective (Mick and Fournier, 1998). We identify consumer control, engagement, and assimilation as three additional media gratifications that consumers derive from multitasking behavior. Further, we argue that multitasking outcomes are paradoxical: Media multitasking may diminish consumers' processing abilities because it can be chaotic,

inefficient, disengaging, and enslaving. Consumers may also be motivated to process advertising in multitasking contexts because of the personal benefits they derive from the process. Thus, marketers should consider designing advertising campaigns that take into account the multitasking benefits and costs identified here. Our study suggests that overcoming the costs associated with media multitasking will enhance the positive impact of media multitasking on MAO.

This study has some limitations that must be considered in evaluating the results. First, the study is exploratory and utilizes qualitative methods to propose a model of the personal outcomes of media multitasking and media multitasking impacts on the MAO model. Thus, we cannot generalize the findings of the study regarding media multitasking behavior. Further, our propositions are based on data derived with a specific sample of a consumer group (Gen Y) that was selected because of its high level of experience with media multitasking. Future research should examine additional consumer segments, including younger segments, such as teens and tweens, as well as older consumers that may exhibit different behaviors and communicate different experiences with respect to media multitasking. While we derive propositions regarding multitasking's impact on the MAO model, future research should investigate empirically the relationships between media multitasking and advertising processing we have identified in this research.

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Appendix: Illustrative Student Collages

Exhibit 1: Student Collage (Steve, 22 years old)

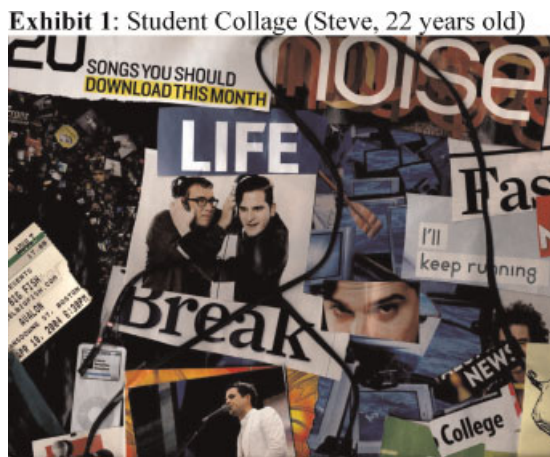


Exhibit 2: Student Collage (Laura, 23 years old)



Exhibit 3: Student Collage (Alex, 20 years old)

